

# How (Not) to Identify Demand Elasticities in Dynamic Asset Markets

van Binsbergen, David, Opp (2026)

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2. Elasticities identified from such shocks cannot be used generically for all counterfactuals (“Can’t clear dynamic markets in state variables”)

# This Discussion

1. Which elasticity is empirically more relevant? Persistent versus transitory demand shocks.
2. Empirical comment on current instruments in DSAP & internal consistency.

## Comment 1a: Definitions

Real question: Which elasticity definition for which counterfactual?

1. This paper ("True Elasticity"): Elasticity to  $\Delta p$  from perfectly **transitory** demand shocks.
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The KY mandate instrument moves quantities persistently, so it identifies the second.

- ▶ Definition 2. is empirically most useful, and clears markets (on average), as I argue next.
- ▶ I view the paper as less of a critique but more as a generalization paired with intuition from an elegant model.

*“How do we formally think about counterfactuals when demand shifts are knowably transitory, identifying the whole spectrum of persistence regimes from 2. to 1.”*

## Comment 1b: Empirical Applicability

What can you do with elasticities identified from persistent demand shifts?

- ▶ Evaluate impact of **persistent** demand shifts – the class from which they were identified

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  - ▶ I totally agree! But *can* clear markets if counterfactual shock has same persistence as instrument. Proof

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In practice demand shocks (and their persistence) are hardly observable.

Investors see prices, and **prices are close to a random walk**.

- ▶ The investor sees a price move and applies an **average persistence**, which is high (knowably transitory shocks are rare).
- ▶ Persistent-shift estimates are arguably the most widely applicable, since the real-world looks persistent from the investor’s view point.

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## Comment 2a: Internal Consistency of KY (2019)

- ▶ “*Can't clear dynamic markets in state variables*”. KY (2019) does not clear markets in state variables.

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- ▶ Explicit Assumption in KY (and other endowment economies):  
Characteristics  $X_t$  are exogenous to prices.<sup>3</sup>
  - ▶ Exactly consistent with the “true” elasticity definition.
  - ▶ One may not like the KY instrument, but the model is internally consistent.

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## Comment 2b: Return Predictability from KY Instrument

- ▶ The paper argues that the KY instrument violates the exclusion restriction by shifting future state variables. Reason for low elasticities.
  - ▶ They show that KY's instrument does not predict returns negatively → No resolution
  - ▶ Very surprising result! **KY instrument is highly correlated with market/book ratio  $mb_t$ .**

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  - ▶ Very surprising result! **KY instrument is highly correlated with market/book ratio  $mb_t$ .**
- ▶ Present-value identity (Cohen Polk Vuolteenaho 2003): controlling for earnings growth,  $mb_t$  predicts future cumulative returns with coeff  $-1$ .

$$mb_t = \sum_{s=1}^{\infty} \rho^s (\mathbb{E}_t[e_{t+s}] - \mathbb{E}_t[r_{t+s}])$$

- ▶ So the instrument should predict negative long-run returns.  
(as long as the additional controls capture earnings growth predictability)

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Standard FMB predictability regression:

$$r_{t+h} = \beta \widehat{me}_t + \gamma be_t + \text{Controls} + \epsilon_t$$

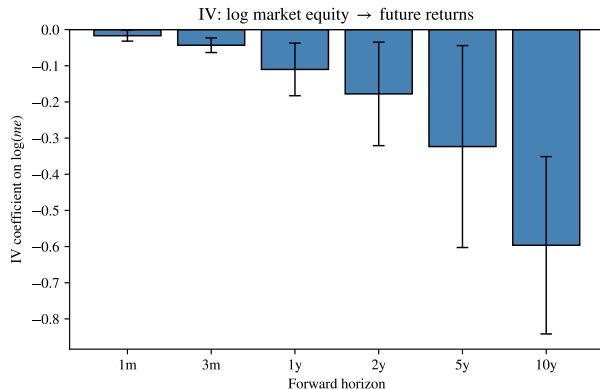
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- ▶ Significantly negative over longer horizons  $h \geq 3m$ .
- ▶ It is the resolution horizon, not instrument validity.
- ▶ The resolution shows up in the long run, not over the next day. (Precisely the trading horizon of long-term buy and hold investors)

Robustness: ex. microcaps

## Comment 2c: Upward-Sloping Demand Curves.

$$\log w(n) = \beta_0 \widehat{me}(n) + \text{Controls} + \epsilon(n)$$

- ▶ For over 90% of institutions we have  $\beta_0 < 1$ . ✓
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→ These are constrained to have  $\beta_0 = 1$  (perfectly passive).
- ▶ Question: What should Blackrock's elasticity be instead?  
→ Does Blackrock provide liquidity to transitory/persistent demand shifts?

# Conclusion

- ▶ **Main point:** An elasticity identified from a shock with persistence  $\rho$  should only be used for counterfactuals with similar persistence. Otherwise, get very strong bias. ✓
  1. Price impact from index inclusions says little about bid-ask spreads.
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  3. ...
- ▶ The paper formalizes this important point in a very nice model.
  - ▶ I learned a lot thinking about this paper and in preparing the discussion!

# Can clear markets with persistent instruments

**Setup.** Demand shock with persistence  $\rho \in [0, 1]$  and linear price impact  $\mathcal{M}$ :

$$z_{t+1} = \rho z_t + \nu_{t+1}, \quad p_t = \mathcal{M} z_t \implies \mathbb{E}_t[p_{t+1}] = \rho p_t$$

**Investor Demand.** True elasticity  $\varepsilon$ , response to future prices  $\kappa$ :

$$q_t = -\zeta p_t + \kappa \mathbb{E}_t[p_{t+1}] \implies q_t = (-\varepsilon + \kappa \rho) p_t$$

**Persistence-conditional elasticity**  $\zeta(\rho)$  not a strictly structural object.

$$\zeta(\rho) = \varepsilon - \kappa \rho$$

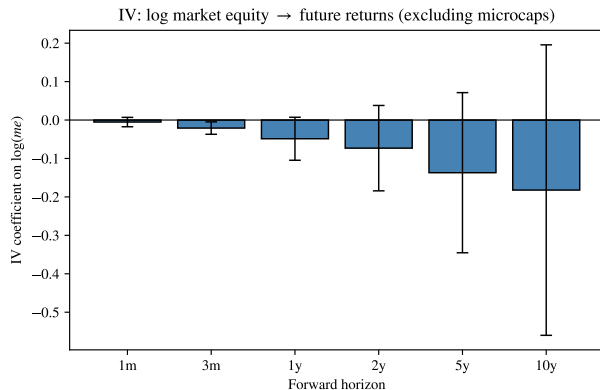
**Equilibrium prices** Measured elasticity  $\zeta(\rho)$  clears the market!

$$p_t = \frac{1}{\zeta(\rho)} \cdot z_t$$

... But the price impact does not identify the structural elasticity  $\varepsilon$ .

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## Appendix: Comment 2 excluding microcaps



- ▶ Same specification, dropping stocks below the NYSE 10th-pct of market cap each quarter.
- ▶ Within-sample rank-standardization.
- ▶ Same horizon profile, so not driven by microcaps.

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